Andrew M. Jones

San Luis Obispo, CA • (530) 206-8704 • andrewmj@outlook.com

Robotics Systems Engineer focused on embedded systems integration, real-time control, and hardware/software debugging. Experienced in optimizing system latency, developing multi-microcontroller architectures, and validating performance through first-principles analysis.

EDUCATION

M.S, B.S in Mechanical Engineering (concurrent), Expected Graduation: June 2026
California State Polytechnic University, San Luis Obispo, CA, Current GPA: 3.9

SKILLS

Embedded & Systems: Embedded Systems, Real-Time Control, STM32, Raspberry Pi, ESP32, Microcontrollers Hardware/Software Integration, System Debugging (Oscilloscope, Signal Analysis)
Programming: Python, MicroPython, C (working knowledge), MATLAB
Robotics & Perception: ROS2, OpenCV, YOLO (Ultralytics), Sensor Integration (IMU, LiDAR, encoders)
Modeling & Analysis: MATLAB (system dynamics), Data Visualization (Matplotlib, Tkinter), System Requirements, Latency Analysis, Failure Mode Analysis (FMEA)

PROJECTS

Robot Visual Tracking and Communication System (Master's Thesis Project – In Progress)
- Designed a multi-microcontroller communication architecture (Raspberry Pi → ESP32 → STM32) to enable real-time perception-to-control data flow
- Defined system latency requirements (<30 ms target) through first-principles analysis of control constraints
- Implemented and profiled SPI, UART, and ESP-NOW protocols to optimize end-to-end latency (current ~50 ms)
- Identified system bottlenecks in image capture and processing pipeline; actively developing optimization strategies to meet real-time constraints
- Built Python-based real-time visualization tools for system performance monitoring and debugging

Autonomous Mobile Robot (Portfolio)
- Developed a real-time embedded control system (MicroPython, STM32) integrating IMU, encoders, and IR sensors
- Designed observer-based state estimator and model-based control to achieve stable autonomous navigation
- Optimized control loop timing to consistently meet 25 ms constraint, enabling reliable estimator performance
- Reduced course completion time from ~30 s to ~15 s through iterative profiling and control algorithm optimization (fastest of 12 teams)
- Built Bluetooth telemetry system to stream pose data and generate plots for rapid debugging and tuning

3-DOF Robotic Harvester System
- Designed and implemented embedded control system for 5-motor robotic manipulator with trajectory planning and safety features
- Debugged motor control instability using oscilloscope analysis of PWM signals, identifying timing inaccuracies from software-generated PWM
- Resolved issue by implementing DMA-based PWM generation, improving signal stability and controller performance
- Performed kinematic modeling (DH parameters) and control tuning to achieve reliable operation

Battle Bot Team Lead
- Led system-level design and integration of a 250-lb combat robot across mechanical and electrical subsystems
- Conducted trade studies evaluating drivetrain and weapon architectures under performance and reliability constraints
- Help craft team vision, develop action items, and ensure execution to achieve project deliverables.
- Furthered design for manufacturability knowledge by CADing and then machining complex robot components using a variety of manufacturing processes.

Autonomous Vehicle Programming (AI)
- Designed and implemented code on scale model autonomous vehicles to achieve various objectives.
- Trained and evaluated YOLO models for object recognition and mask generation. Deployed in ROS2 python environment to implement line following and sign recognition
- Tested and built code in simulated environments before deployment
- Worked with a variety of sensors including LiDAR and ultrasonic sensors

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WORK EXPERIENCE

Air Treatment Corporation (Internship) Rancho Cordova, CA
Sales Engineering Intern Jun 2025 – Sep 2025

- Increased knowledge of sales engineering processes by supporting both inside and outside sales with product selections and scope validation through job walks and customer meetings
- Increased understanding of commercial HVAC systems and design criteria including air quality, equipment capacity, efficiency, and regulatory compliance
- Completed comprehensive evaluation of existing office HVAC system. Compared in-kind and modernized replacement options. Evaluated first-costs and operating savings. Presented recommendations to leadership

Northen California Power Agency (Internship) Murphys, CA
Hydroelectric Plant Intern Jun 2024 – Sep 2024

- Assisted Chief Dam Safety Engineer in ensuring regulatory compliance of complex hydroelectric projects
- Increased mechanical skills by assisting with turbine and dam maintenance and repair
- Refined communication skills by writing requests for proposals and bid documents
- Increased interdisciplinary skills by working with engineers in wilderness environments

AirPoint Precision Inc. Diamond Springs, CA
Machine Operator Jan 2023 – Jul 2023

- Operated CNC mills and lathes; machined parts for a wide variety of applications
- Enhanced ability to read and interpret engineering drawings
- Practiced quality control by using various digital and analog measuring devices to inspect parts

OmniCubed Inc. Shingle Springs, CA
Material Handler Jan 2021 – Aug 2022

- Worked as part of a team assembling stone installation tools from CNC machined and injected molded parts
- Improved ability to dynamically adjust to situations by assisting shipping and receiving department and machine shop as needed